



FOOD INVENTORY MANAGEMENT PLATFORM FOR SMBS

Database Foundations



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PROJECT SCOPE

- Target: Small F&B businesses
- Main Goal: Reduce food waste + organize the inventory + reduce cost



THE PAIN POINTS & SOLUTION

Pain Points:

- Excessive Waste: Ingredients expire unnoticed.
- Inventory Chaos: Running out of stock during peak hours.
- Supplier Issues: Supplier choices are limited.

The Solution:

- Automated Expiry Alerts.
- FIFO (First-In-First-Out) Inventory Tracking.
- Integrated B2B Marketplace.

Price Model: 15% Commission on Orders + 200 THB Monthly Subscription.

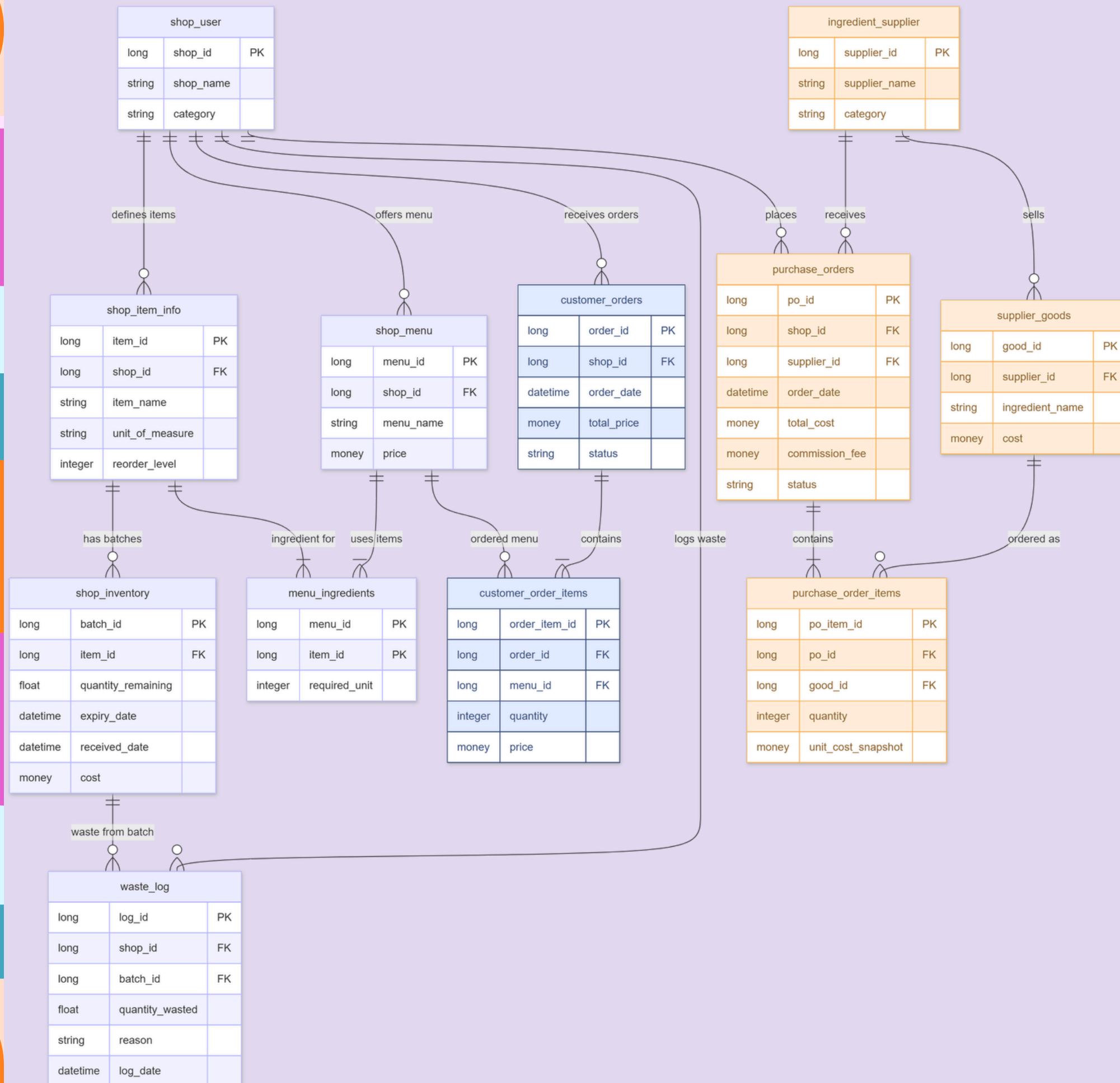
DATABASE DESIGN

5 Key Modules Highlighted:

- 1.Shop & Inventory
- 2.Menu & Recipe
- 3.Sales (POS)
- 4.Supplier Market
- 5.Waste Log

3NF achieved:

- No repeating groups
- No partial dependencies
- Every non-key depends on the key



SECURITY & DATA INTEGRITY IMPLEMENTATION

1.Security Implementation

Password Hashing: To protect data from unauthorized access, the shop_user table stores a password_hash. This ensures that even if the database is compromised, user credentials remain secure.

	shop_id [PK] bigint	shop_name character varying (255)	category character varying (100)	is_active boolean	password_hash character varying (255)
1	1	The Tasty Table	Restaurant	true	\$2kueThisIsAFakeHashString
2	2	Yummy Yum	Restaurant	true	\$2kueThisIsAFakeHashString

	supplier_id [PK] bigint	supplier_name character varying (255)	category character varying (100)	password_hash character varying (255)
1	1	Global Matcha Co	Tea	\$2kueThisIsAFakeHashStri...
2	2	Dairy Farm	Dairy	\$2kueThisIsAFakeHashStri...

SECURITY & DATA INTEGRITY IMPLEMENTATION

1. Data Integrity Implementation

Soft Deletes: This implementation allows for inactivating a shop without losing data in case the shop user decides not to use the platform anymore

```
ALTER TABLE shop_user ADD COLUMN is_active BOOLEAN DEFAULT TRUE;
```

Time Consistency: The database utilized the TIMESTAMPTZ for date data. This enforces a domain of global time accuracy.

	batch_id [PK] bigint	item_id bigint	quantity_remaining numeric (10,2)	expiry_date timestamp with time zone	received_date timestamp with time zone	cost numeric (10,2)
1	1	1	1000.00	2026-06-01 12:23:53.975786+...	2025-11-28 12:23:53.975786+...	20.00
2	2	2	5000.00	2025-12-10 12:23:53.975786+...	2025-12-02 12:23:53.975786+...	0.05
3	3	3	2000.00	2025-12-13 12:23:53.975786+...	2025-12-02 12:23:53.975786+...	0.08

SECURITY & DATA INTEGRITY IMPLEMENTATION

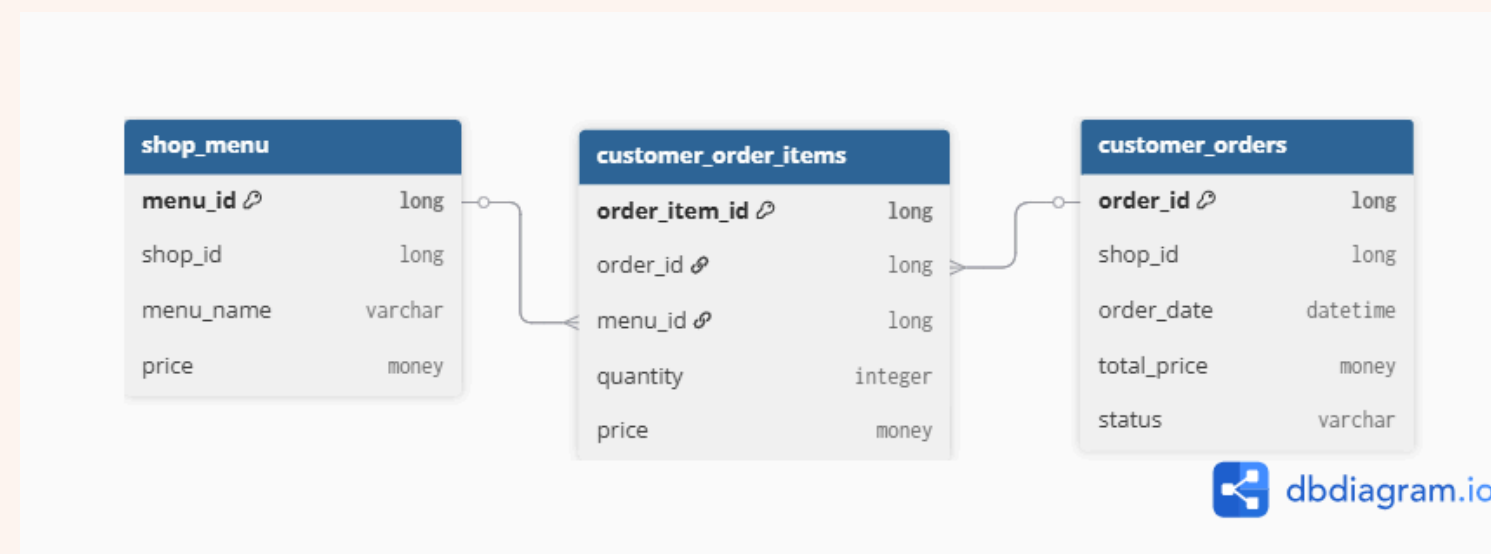
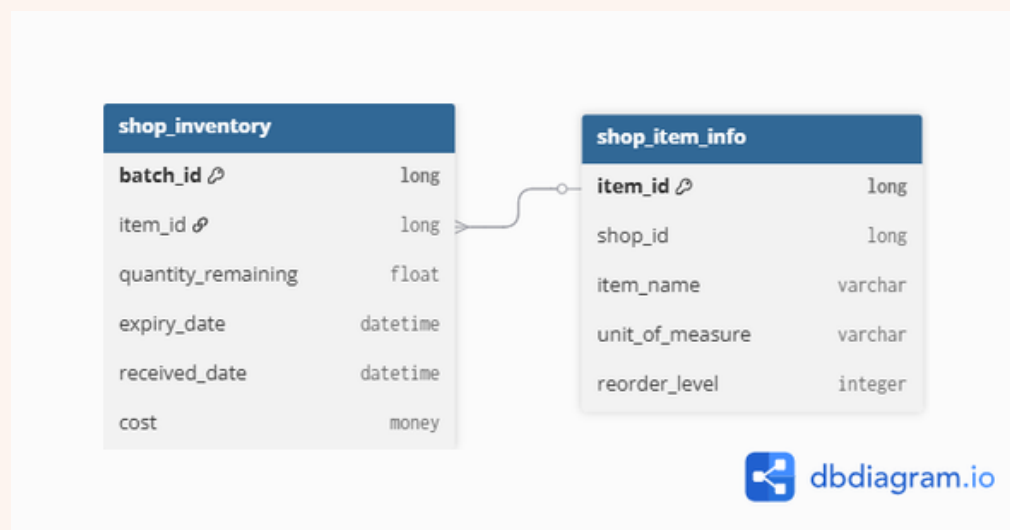
Entity Integrity: Every table in the database has a BIGSERIAL PRIMARY KEY

- The table for entity identifiers enforces NOT NULL to prevent incomplete records from entering the system.

```
CREATE TABLE shop_user (  
  shop_id BIGSERIAL PRIMARY KEY,  
  shop_name VARCHAR(255) NOT NULL,  
  category VARCHAR(100)  
);
```

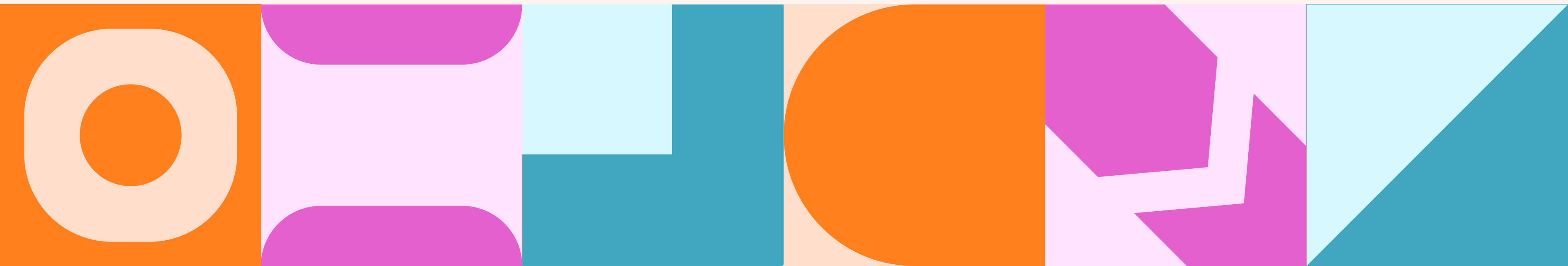
Referential Integrity: The database uses strict FOREIGN KEY References to maintain data consistency.

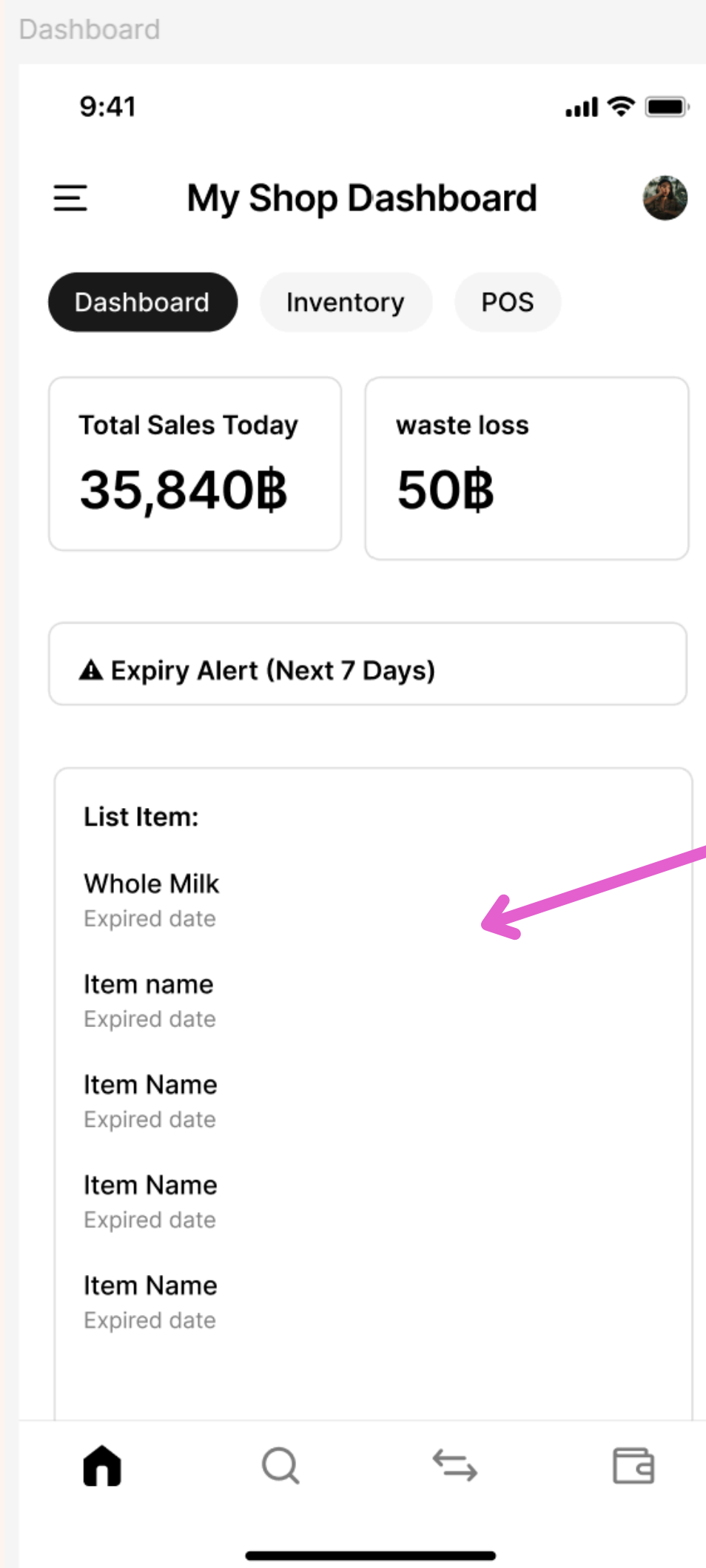
- This ensures that every record must be linked to a valid parent entry.





KEY FEATURES & FUNCTION IMPLEMENTATION





FEATURE A: WASTE ALERT DASHBOARD

Query Query History Scratch Pad X

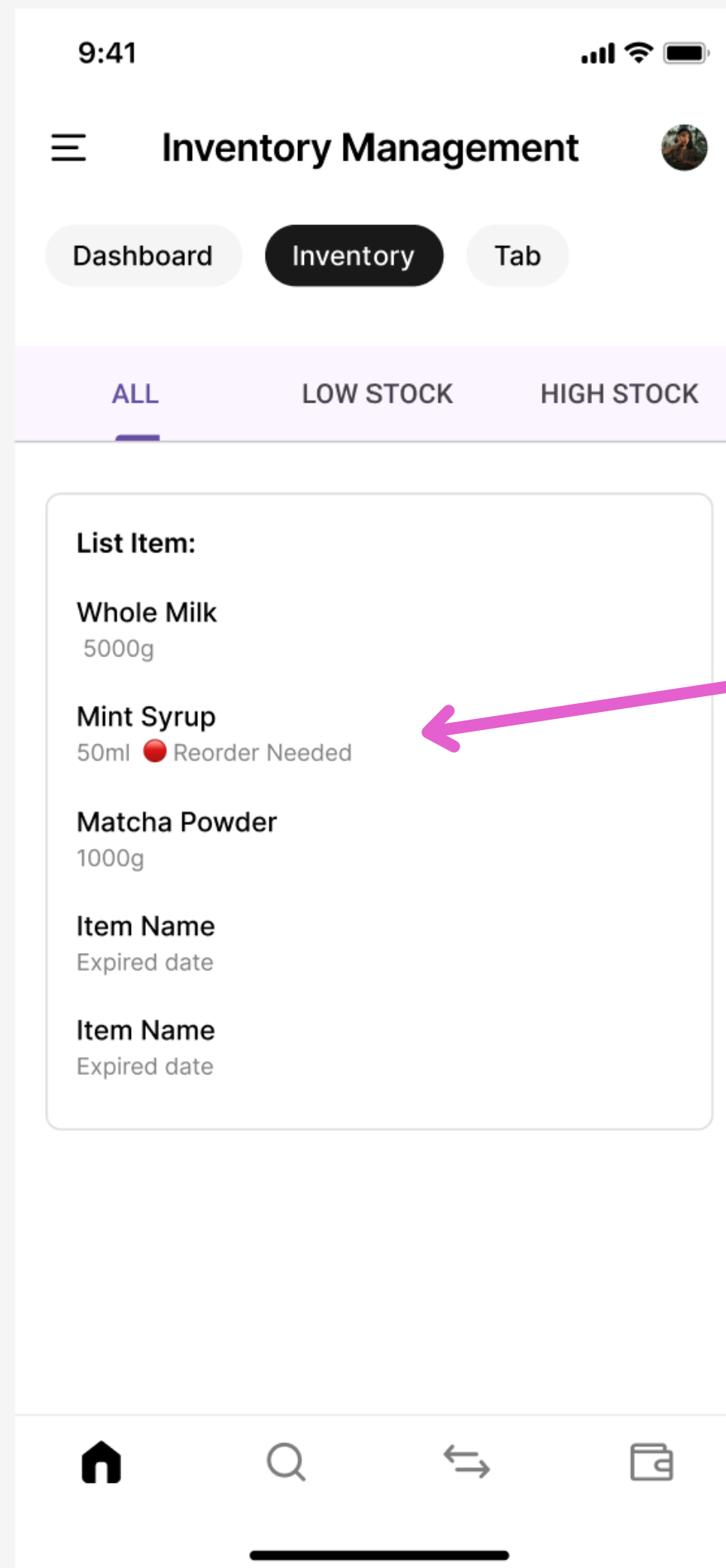
1 `SELECT * FROM check_expiring_inventory(5, 7);`

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1

	batch_id bigint	item_name character varying	expiry_date timestamp with time zone	quantity_remaining numeric	days_until_expiry integer
1	2	Whole Milk	2025-12-10 12:23:53.975786+...	5000.00	6

- Function Call:
`SELECT * FROM check_expiring_inventory(5, 7);`
- The shop owner checks the dashboard to see expiring items in 7 days to prevent food waste

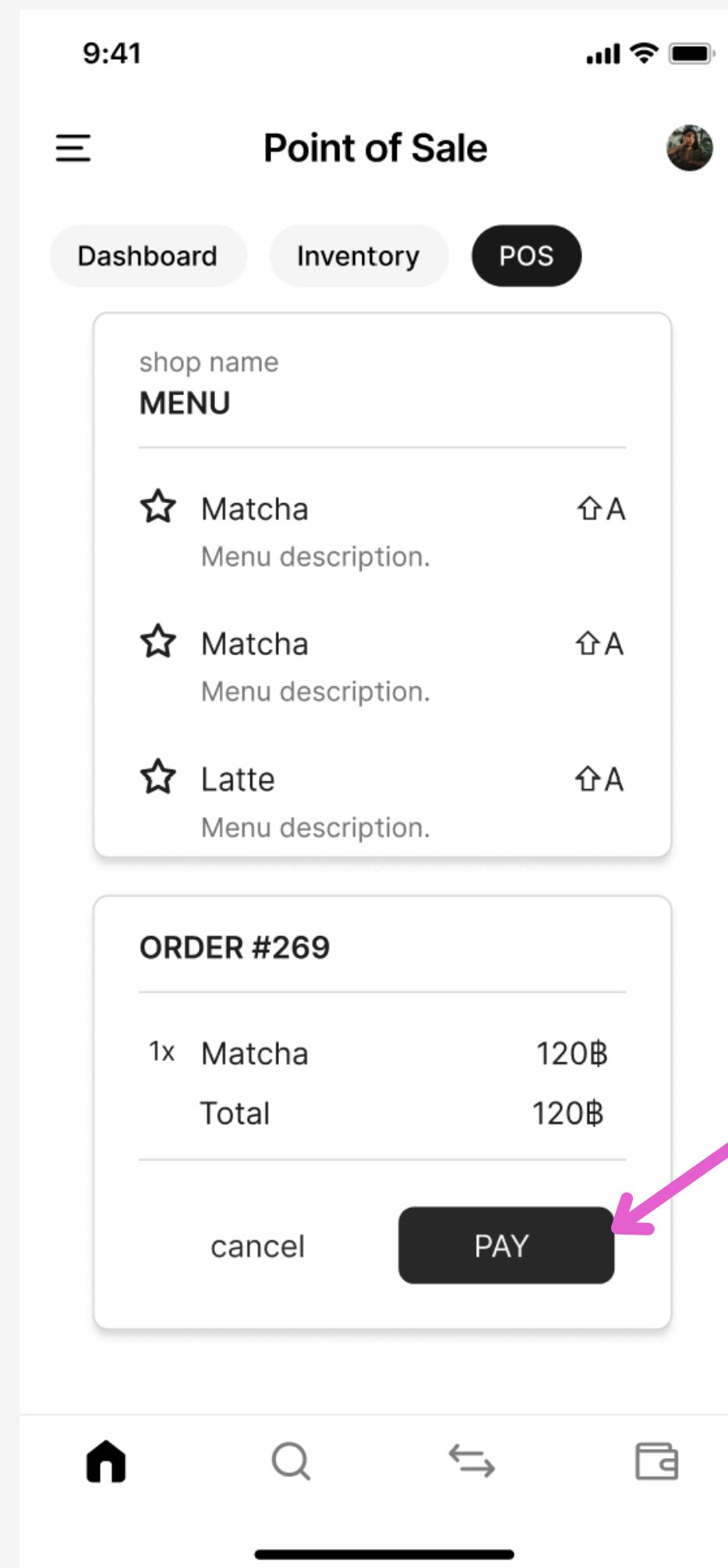


FEATURE B: RESTOCK ALERT

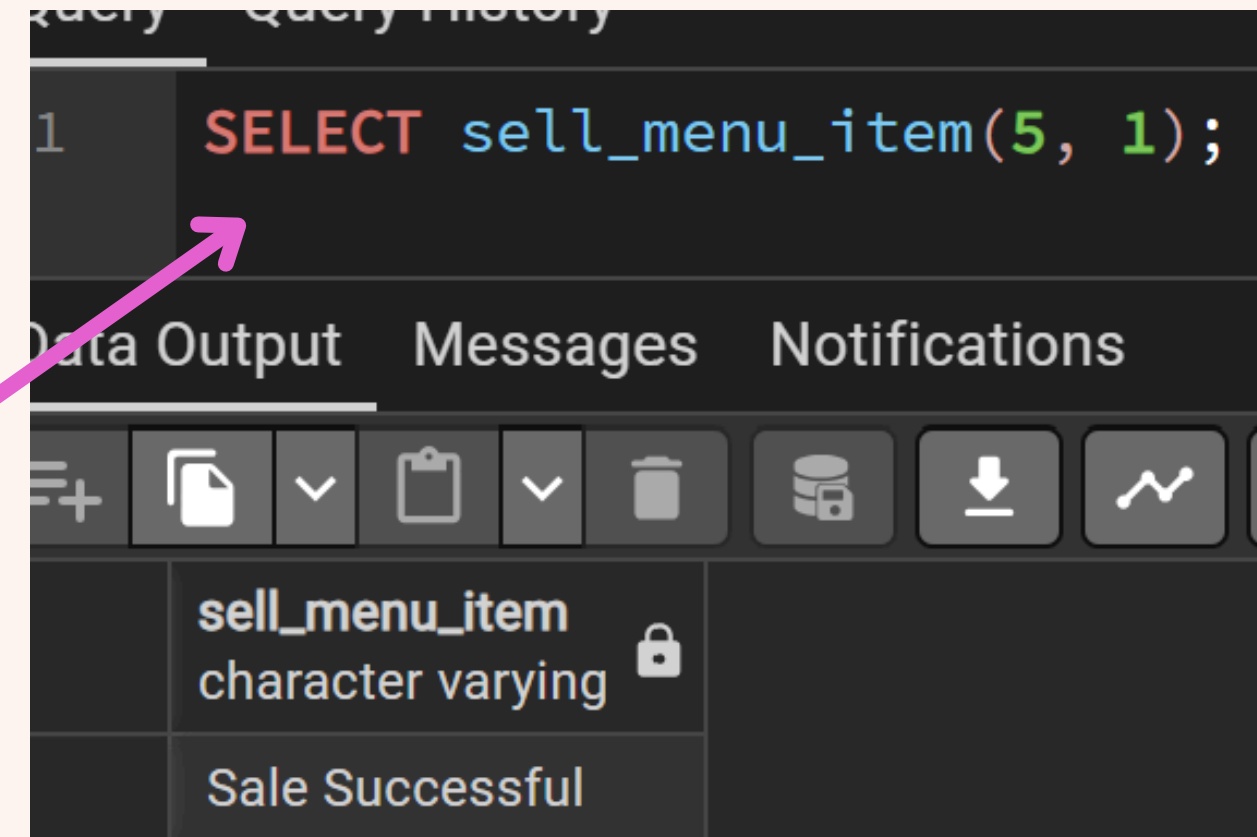
The screenshot shows a SQL query interface. At the top, there's a 'Query History' section. Below it, a query is entered: `SELECT * FROM check_restock_level(5);`. Below the query, there's a 'Data Output' section with tabs for 'Data Output', 'Messages', and 'Notifications'. The 'Data Output' tab is selected, showing a table with the following data:

item_name	total_quantity	reorder_level	status
character varying	numeric	integer	character varying
Mint Syrup	50.00	500	Restock Needed
Brown Sugar	0	100	Restock Needed
Simple Syrup	150.00	200	Restock Needed

- Function Call: `SELECT * FROM check_restock_level(5);`
- The shop manager checks inventory to see what needs to be reordered.



FEATURE C: POINT OF SALE



- A staff member processes a customer sale.
- Function Call: `SELECT sell_menu_item(5, 5);`
- The function looks up the recipe, then finds the oldest available batch and deducts the ingredient quantity

Inventory before run function

1

SELECT * FROM get_shop_inventory(5);

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 7

	item_name character varying	batch_id bigint	quantity_remaining numeric	expiry_date timestamp with time zone
1	Matcha Powder	1	1000.00	2026-06-01 12:23:53.975786+...
2	Whole Milk	2	5000.00	2025-12-10 12:23:53.975786+...
3	Oat Milk	3	2000.00	2025-12-13 12:23:53.975786+...
4	Cocoa Powder	4	2000.00	2026-12-03 12:23:53.975786+...
5	Mint Syrup	5	50.00	2026-03-03 12:23:53.975786+...
6	Arabica Beans	6	3000.00	2026-01-02 12:23:53.975786+...
7	Simple Syrup	7	150.00	2026-12-03 13:05:01.770595+...

Inventory after run function

SELECT * FROM get_shop_inventory(5);

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to 7

	item_name character varying	batch_id bigint	quantity_remaining numeric	expiry_date timestamp with time zone
	Oat Milk	3	2000.00	2025-12-13 12:23:53.975786+...
	Cocoa Powder	4	2000.00	2026-12-03 12:23:53.975786+...
	Mint Syrup	5	50.00	2026-03-03 12:23:53.975786+...
	Arabica Beans	6	3000.00	2026-01-02 12:23:53.975786+...
	Matcha Powder	1	995.00	2026-06-01 12:23:53.975786+...
	Whole Milk	2	4850.00	2025-12-10 12:23:53.975786+...
	Simple Syrup	7	130.00	2026-12-03 13:05:01.770595+...

9:41

<

Checkout

SHIPPINGID

#158756

>

Free

Standard | 3-4 days

>

PAYMENT

Visa *1234

>

PROMOS

Apply promo code

>

ITEMS



Brand

Product name

Description

Quantity: 02



Global Matcha Co.

Premium Matcha 1kg

Description

Quantity: 01

PRICE

1000฿

500฿

Subtotal (3)

2500฿

Shipping total

FREE

Total

2500฿

Place order

FEATURE D: SUPPLIER MARKETPLACE

- Function Call: SELECT start_purchase_order(5, 1);
- When shop_user starts placing an order, the system creates a "Draft" order

```
1 SELECT start_purchase_order(5, 1);
```

	Data Output	Messages	Notifications
	start_purchase_order bigint		
1			12

```
1 SELECT add_purchase_item(12, 1, 1, 500.00);
2 SELECT add_purchase_item(12, 2, 2, 1000.00);
```

Add items to purchase order

```
1 SELECT finalize_purchase_order(12);
```

Query	Query History	Scratch Pad	X
1	SELECT * FROM purchase_orders WHERE po_id = 12;		
	Data Output	Messages	Notifications
	Showing rows: 1 to 1	Page No: 1	of 1
	po_id [PK] bigint	shop_id bigint	supplier_id bigint
1	12	5	1
	order_date timestamp with time zone	total_amount numeric (10,2)	commission_fee numeric (10,2)
	2025-12-03 22:13:40.662108+...	2500.00	375.00
			status character varying (50)
			ordered

- Function Call: SELECT finalize_purchase_order(12);
- This function sums the items, calculates the 15% Commission, and updates the status to 'ordered'.

9:30

Record Waste

Select Item:

Whole Milk (2L)

Select Batch:

Batch #102 (Exp: 2025-12-05)

Reason for Waste:

Spilled by Staff

Quantity Wasted

500

CONFIRM WASTE

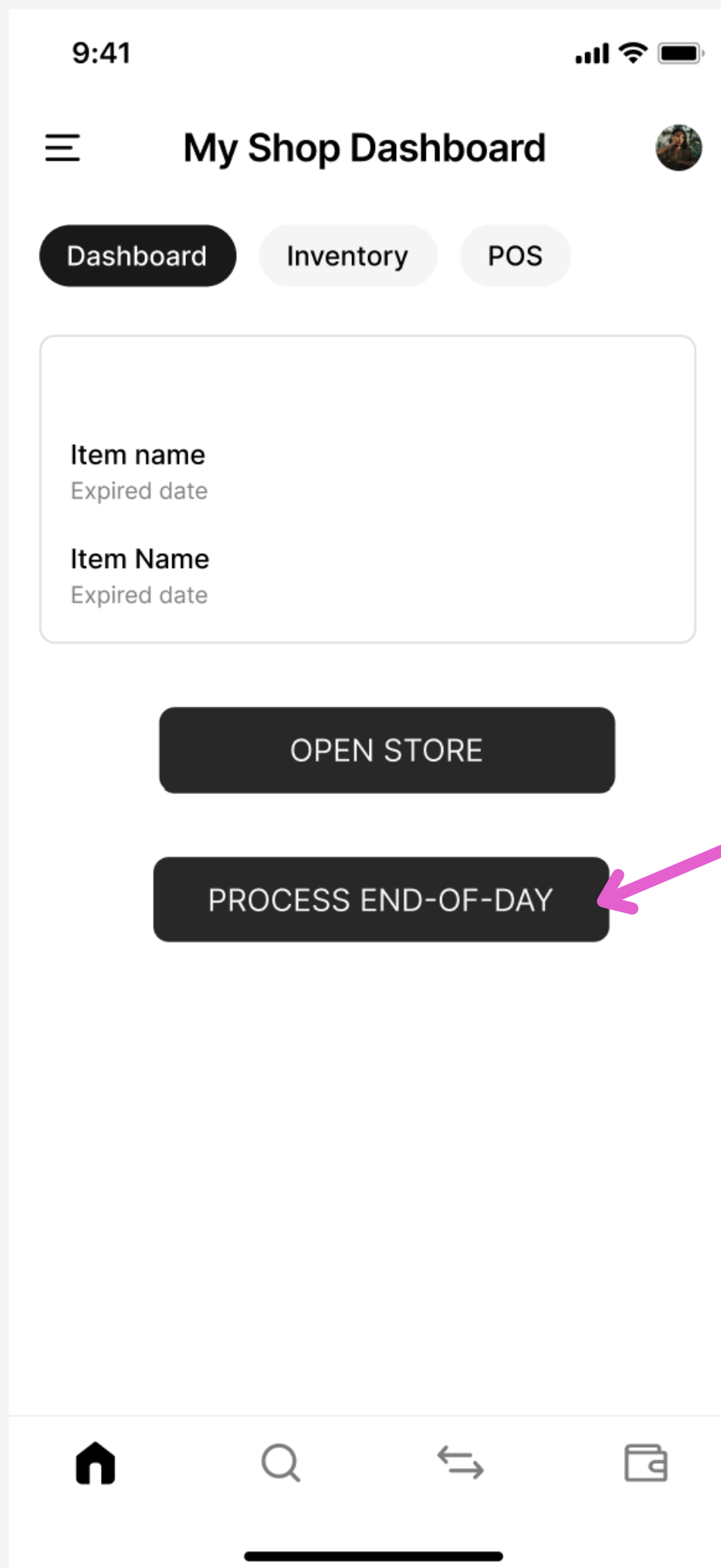
FEATURE E: MANUAL WASTE RECORDING

```
SELECT record_waste(2, 10, 20, 'Spilled');
```

```
1 SELECT * FROM waste_log WHERE shop_id = 2;
```

Data Output Messages Notifications							
Showing rows: 1 to 5 Page No: 1							
	log_id [PK] bigint	shop_id bigint	batch_id bigint	quantity_wasted numeric (10,2)	reason character varying (255)	log_date timestamp with time zone	
1	1	2	17	5.00	Dropped	2025-12-03 21:46:23.289355+...	
2	2	2	16	2.00	Expired	2025-12-02 21:46:23.289355+...	
3	7	2	15	1.00	Sample Testing	2025-12-03 21:46:23.289355+...	
4	8	2	14	10.00	Expired	2025-12-01 21:46:23.289355+...	
5	9	2	10	20.00	Spilled	2025-12-03 22:34:31.27878+00	

- Adjust stock when the ingredients are immediately wasted. For example, the staff accidentally spills milk.
- The function targets that specific batch, moves the quantity to the waste_log table, and updates the inventory to the current quantity.



FEATURE F: ADMIN END-OF-DAY PROCESS

```
SELECT process_daily_expiry(5);
```

- Function Call: `SELECT process_daily_expiry(5);`
- The shop manager closes the shop and ensures expired items are cleared from the active inventory and move to `wast_log`.
- The function loops through the entire inventory. It finds any batch where expiry date less than today date, logs it as waste, and clears the stock automatically.

DELIVERABLES & ACCESS

GitHub Repository: <https://github.com/Hamtaro-chan/fnb-inventory-management-platform>

Live Database Connection (Render):

Hostname: dpg-d4jklvndiees738rohfo-a

Port: 5432

Database: fnb_platform_db

Username: root

Password: X7pbXxNI5B0UdqpSKjQKctlkubeuFEGm

External Database URL:

postgresql://root:X7pbXxNI5B0UdqpSKjQKctlkubeuFEGm@dpg-d4jklvndiees738rohfo-a.singapore-postgres.render.com/fnb_platform_db

CONCLUSIONS

- Solves key F&B pain points
- Secure, normalized, and scalable database
- Practical PL/pgSQL features
- Supports real-world operations
- Fully deployed and tested



THANK YOU
SO MUCH

